WEEK 1 INTERNSHIP REPORT

Python for Machine Learning & Data Preprocessing

Internship: YuvaIntern — AI Pioneers Internship

Task: Week 1 — Python for Machine Learning & Data Preprocessing

Student name: [Enter your name]

College: [Enter your college]

Submission date: [Enter date]

GitHub repository: [Paste your repository URL]

1. Abstract

This project demonstrates a beginner-friendly data-preprocessing workflow using the Titanic passenger dataset. Python libraries were used to inspect the dataset, identify missing values, clean data, encode categorical features, scale numerical variables, perform exploratory data analysis (EDA), and export prepared datasets.

2. Objectives

Understand Python-based data preparation; load and inspect CSV data; detect missing values and duplicates; apply missing-value strategies; encode categorical features and scale numerical features; select candidate features; perform EDA; document the process.

3. Dataset description

The public Titanic passenger training dataset contains passenger information such as class, sex, age, family counts, fare, embarkation port, and survival status. Survived is retained as the target label (0 = did not survive, 1 = survived). The original data is preserved separately.

4. Tools and libraries

Python, Pandas, NumPy, Matplotlib, Seaborn, and scikit-learn.

5. Methodology

Data inspection: checked shape, data types, summary statistics, duplicates, and missing values. Missing values: filled Age using the median and Embarked using the mode. Cabin was dropped from the working feature table because many entries are missing. Duplicate rows were removed from the working copy. Feature selection: kept Pclass, Sex, Age, SibSp, Parch, Fare, and Embarked as candidate inputs, with Survived as target. PassengerId, Name, Ticket, and Cabin were not used as model features. Categorical encoding: one-hot encoded Sex and Embarked. Scaling: standardized Age and Fare using StandardScaler. For a future model, fit the scaler only on training data to avoid leakage.

6. Results

Run the notebook in Google Colab and replace the placeholders in the table below with the actual values printed in outputs/results_summary.txt. Do not submit placeholders.

Measure	Actual result (fill after running notebook)
Original dataset dimensions	[rows] × [columns]
Duplicate rows found / removed	[value] / [value]
Missing cells before / after	[value] / [value]

Measure	Actual result (fill after running notebook)
Median used for Age	[value]
Mode used for Embarked	[value]
Cleaned dataset dimensions	[rows] × [columns]
Encoded matrix dimensions	[rows] × [columns]
Overall survival rate	[value]%

7. EDA observations

After running the notebook, write 2–4 observations from your actual charts and tables. Describe patterns only; do not claim causation.

8. Output files

The notebook creates original, cleaned, encoded, and scaled CSV files, an EDA chart image, and a results_summary.txt file.

9. Conclusion

This task provides practical experience with missing-value handling, duplicate removal, categorical encoding, numerical scaling, feature selection, and EDA. The outputs form a foundation for later classification work.

10. References

Kaggle Titanic dataset: https://www.kaggle.com/competitions/titanic/data ; Pandas tutorials: https://pandas.pydata.org/docs/getting_started/intro_tutorials/ ; scikit-learn preprocessing: https://scikit-learn.org/stable/modules/preprocessing.html ; Google Colab: https://colab.research.google.com/